LOS Assignment 2

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1.Compare Permissions on Regular Files and Directories

Create a regular file named sample.txt and a directory named sample_dir. Create at least one file inside sample_dir.
Apply each of the following permissions using chmod. Perform the specified operations and record your observations.

Permission	Regular File (sample.txt)	Directory (sample_dir)
400 (r--)	The file can be read, but it cannot be modified because there is no write permission.	The directory can be listed, but it cannot be entered/accessed because there is no execute permission.
200 (-w-)	The file cannot normally be read, but its contents can be overwritten or changed if a command does not need to read the old contents.	The directory cannot be listed or entered, and a new file cannot be created because execute permission is missing.
100 (--x)	The file cannot be read. It can be executed only if it is actually an executable file/script and the system allows it to run	The directory cannot be listed, but it can be entered/accessed if you know the directory name and have the necessary path permissions.
500 (r-x)	The file can be read and executed, but it cannot be modified because there is no write permission.	The directory can be listed and entered, but files cannot be created or deleted because there is no write permission.
600 (rw-)	The file can be read and modified, but it cannot be executed.	The directory can be read/listed, but it cannot be entered or have files created because execute permission is missing.
700 (rwx)	The owner can read, modify, and execute the file. This gives full permission to the file.	The owner can list and enter the directory, create files, and delete files inside it.

Questions and Answers

1. What does read (r) permission allow you to do with a regular file?
Read permission allows you to view the contents of the file. For example, you can use cat sample.txt to see what is inside it.

2. What does read (r) permission allow you to do with a directory?
For a directory, read permission allows you to see the names of the files and subdirectories inside it. In other words, it allows you to list the directory contents.

3. What does write (w) permission mean for a regular file?
Write permission allows you to change the contents of the file. You can add, remove, or replace data in the file.

4. What does write (w) permission mean for a directory?
Write permission on a directory allows you to add, remove, or rename entries inside the directory. However, in practice, you also need execute (x) permission on the directory to actually perform these operations.

5. What does execute (x) permission mean for a regular file?
For a regular file, execute permission means the operating system allows the file to be run as a program or script, assuming its contents are something executable.

6. Why is execute (x) permission important for accessing a directory?
Execute permission on a directory means you are allowed to enter the directory and access items inside it. Without it, you may be able to see the names of files with read permission, but you generally cannot actually access those files through the directory.

7. Which permission combination allows the owner to fully access a regular file?
700 (rwx) gives the owner full permissions: read, write, and execute.

8. Which permission combination allows the owner to fully access a directory?
700 (rwx) gives the owner full permissions to list the directory, enter it, create files, delete files, and manage its contents.

9. Based on your observations, explain why the same r, w, and x permissions behave differently for a regular file and a directory.
The main reason is that a file and a directory are different types of objects. For a regular file, the permissions apply directly to the file's contents: r means reading the contents, w means changing the contents, and x means running the file.